

6 Days TAT – Project Execution Plan

Day 1 – Problem Framing & System Architecture

Objective: Think like a decision owner, not a model

builder. **Work Done:**

- Deeply understand the **business risk** of blind ML decisions •

Finalize **decision ownership philosophy** → *System > ML* •

Define:

- Confidence zones (HIGH / MEDIUM / LOW)
- Risk dimensions (amount, frequency, historical behavior)
- Design **end-to-end architecture**:
 - ML model (input only)
 - Decision Control Engine
 - Override & escalation flow
- Create high-level flow diagram (ML → Risk → Decision → Explanation)

Deliverables:

- Final problem statement
- Decision flow logic
- Architecture blueprint

Day 2 – Data Design & Risk Feature Engineering

Objective: Convert raw data into decision-ready signals.

Work Done:

- Design MySQL schema for:
 - Transaction events
 - Decision logs
 - Human review flags
- Identify **risk features**:
 - Transaction amount risk
 - Behavioral spike indicators
 - User reliability score

- Define thresholds (example):
 - High-value transaction limit
 - Frequency anomaly boundaries
- Prepare clean feature extraction logic

Deliverables:

- Database schema
- Risk feature definitions
- SQL joins & extraction logic

Day 3 – ML Model Integration (Not Control)

Objective: ML predicts — but does not decide.

Work Done:

- Train / integrate ML model for base prediction
- Output only:
 - Prediction (ALLOW / REJECT)
 - Probability score
- Validate model behavior on edge cases
- Ensure **model confidence is treated as an input**, not final truth
- Store ML outputs for audit (no direct execution)

Deliverables:

- Trained ML model
- Prediction + probability output structure
- ML evaluation metrics

Day 4 – Decision Override & Risk Control Engine

Objective: Build the brain of the system.

Work Done:

- Implement **confidence zone classifier**
- Build **rule-based override logic**:
 - ML approve + high risk → override

- Medium confidence → escalate
- Low confidence → auto reject / manual
- Implement **Human-in-the-Loop trigger** • Ensure

ML disagreement is allowed by design **Deliverables:**

- Decision Control Engine code
- Override rules
- Escalation logic

Day 5 – Explainability & REST API Development

Objective: Make decisions trustworthy and auditable. **Work Done:**

- Build **Explanation Generator:**
 - Why ML was trusted / overridden
 - Which risks triggered the decision
- Develop FastAPI endpoint:
 - POST /decision/evaluate
- Standardize API responses:
 - Final decision
 - ML output
 - Confidence
 - Decision path
- Log every decision for compliance

Deliverables:

- REST API implementation
- Explainable decision response
- Decision audit logs

Day 6 – Testing, Validation & Production

Readiness Objective: Ensure enterprise-grade reliability. **Work Done:**

- Test APIs using Postman:
 - High-risk scenarios
 - Low-confidence cases
 - Override validations
- Simulate real failure cases
- Validate human escalation flow •

Finalize documentation:

- System guarantees
- Decision responsibility
- Failure handling strategy

Deliverables:

- Postman test collection
- Final project documentation •

Production-ready decision system